NON-STAB POWER OF FOURIER TRANSFORM

- WE HAVE NUMERICAL EVIDENCE THAT QUANTUM FOURIER TRANSFORM IS MAGIC FULL
- HOW TO ANALYTICALLY DO IT?

$$M_{LIN}(U) = \frac{d}{2(4+d)\,\mathrm{Tr}(Q\Pi_{SYM})} \left\| [U^{\otimes 4}, Q\Pi_{SYM}] \right\|_2^2$$

IFF $[U^{\otimes 4}, Q\Pi_{SYM}] = 0$

THE UNITARY INTRODUCES MAGIC

- WE KNOW

$$\mathrm{Tr}(Q\Pi_{SYM}) = \frac{(d+1)(d+2)}{6}$$

$$Q = \frac{1}{d^2} \sum_P P^{\otimes 4}$$

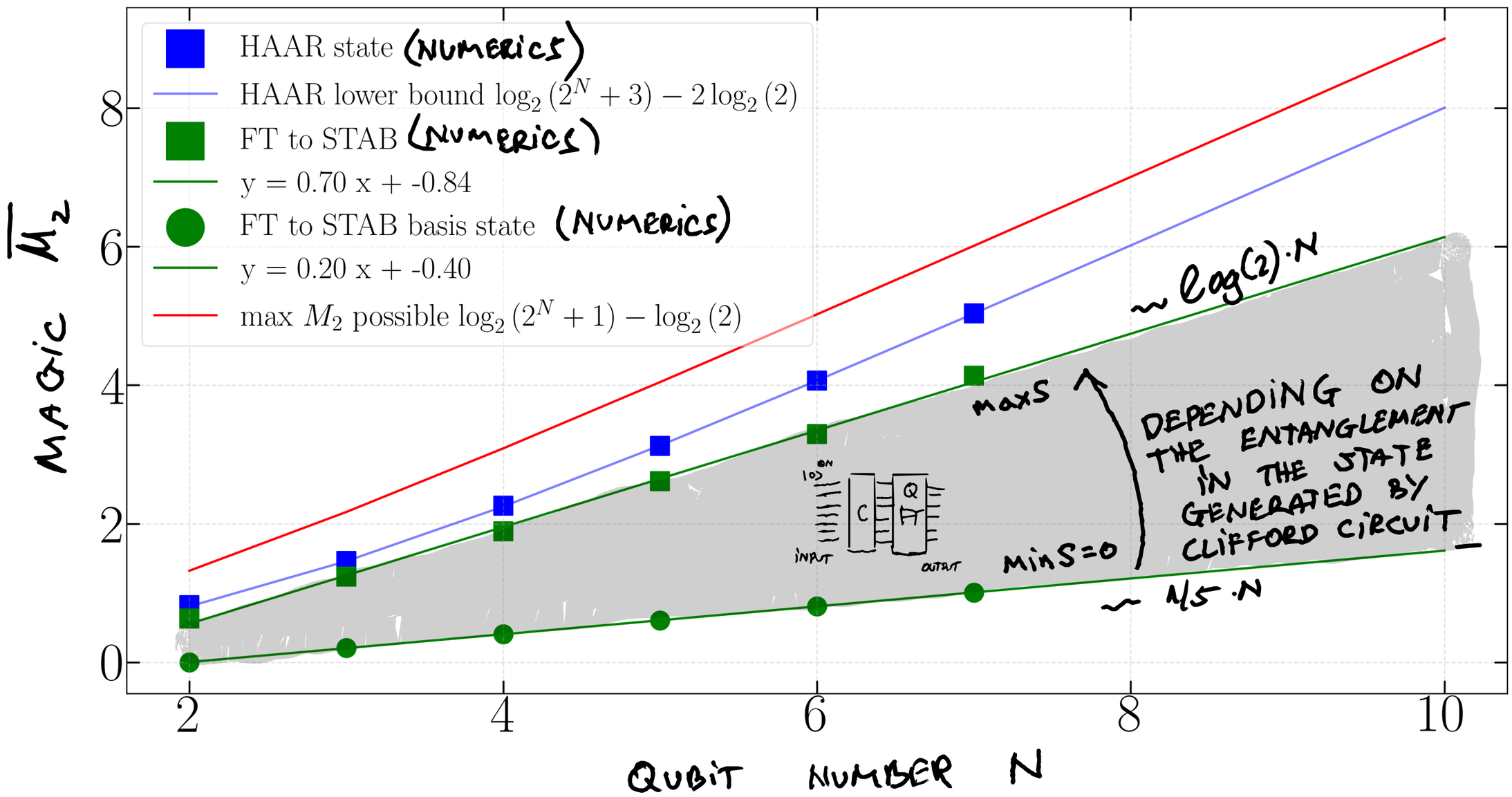
ALSO DIFFERENCE BETWEEN AFTER AND BEFORE

$$\Delta Q\Pi_{SYM} := (U^\dagger)^{\otimes 4} Q\Pi_{SYM} U^{\otimes 4} - Q\Pi_{SYM}$$

WHAT HAPPENS AFTER THE UNITARY EVOLUTION

BEFORE

$$M_{LIN}(U) = \frac{d}{2} \frac{(d+1)(d+2)}{6(d+4)} \| \Delta Q\Pi_{SYM} \|_2^2$$



$$\bar{M}_2(U^{FT}) = \mathbb{E}_C\left[M_2(U^{FT}C|0\rangle^{\otimes N})\right]$$

$$M_2 = -\log_2\left(\frac{1}{d}\sum_{P\in P_n} Tr^4(PS)\right)$$

OF REALIZATIONS = 100

$$U^{QFT} = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \sum_{k=0}^{N-1} \omega^{ik}(N) |k><j|$$

$$\omega^{jk}(N) = e^{2\pi i \frac{jk}{N}}$$

SUPPRES N NOTATION!

$N = 2^n$

$$\left(U^{QFT}\right)^{\otimes 4} = \left(\frac{1}{\sqrt{N}}\right)^4 \sum_{j_1,k_1=0}^{N-1} \sum_{j_2,k_2=0}^{N-1} \sum_{j_3,k_3=0}^{N-1} \sum_{j_4,k_4=0}^{N-1}$$

$$\omega^{j_1 k_1} \omega^{j_2 k_2} \omega^{j_3 k_3} \omega^{j_4 k_4} |k_1 k_2 k_3 k_4><j_1 j_2 j_3 j_4|$$

WHEN WE APPLY PERMUTATIONS

$$T_{1234}\left(U^{QFT}\right)^{\otimes 4} = \left(U^{QFT}\right)^{\otimes 4} \quad \text{TRIVIAL}$$

$$T_{2134}\left(U^{QFT}\right)^{\otimes 4} = \left(\frac{1}{\sqrt{N}}\right)^4 \sum_{j_1,k_1=0}^{N-1} \sum_{j_2,k_2=0}^{N-1} \sum_{j_3,k_3=0}^{N-1} \sum_{j_4,k_4=0}^{N-1}$$

$$\omega^{j_1 k_1} \omega^{j_2 k_2} \omega^{j_3 k_3} \omega^{j_4 k_4} \; T_{2134} |k_1 k_2 k_3 k_4><j_1 j_2 j_3 j_4|$$

$$= \left(\frac{1}{\sqrt{N}}\right)^4 \sum_{j_1,k_1=0}^{N-1} \sum_{j_2,k_2=0}^{N-1} \sum_{j_3,k_3=0}^{N-1} \sum_{j_4,k_4=0}^{N-1} \omega^{j_1 k_1} \omega^{j_2 k_2} \omega^{j_3 k_3} \omega^{j_4 k_4} \; |k_2 k_1 k_3 k_4><j_1 j_2 j_3 j_4|$$

COMMUTATOR (RELEVANT PART)

$$(U^\dagger)^{\otimes 4} \, Q \, \Pi_{SYM} \, U^{\otimes 4}$$

DEFINITIONS:

$$Q = \frac{1}{d^2} \sum_{P}' P^{\otimes 4}$$

$$\Pi_{SYM} = \frac{1}{4!} \sum_{\pi \in S_4}' T_\pi$$

$$\left((U^{QFT})^\dagger\right)^{\otimes 4} = \left(\frac{1}{\sqrt{N}}\right)^4 \sum_{j_1,k_1=0}^{N-1}{}' \sum_{j_2,k_2=0}^{N-1}{}' \sum_{j_3,k_3=0}^{N-1}{}' \sum_{j_4,k_4=0}^{N-1}{}'$$

$$\left(\omega^{j_1 k_1}\right)^* \left(\omega^{j_2 k_2}\right)^* \left(\omega^{j_3 k_3}\right)^* \left(\omega^{j_4 k_4}\right)^* |k_1 k_2 k_3 k_4\rangle\langle j_1 j_2 j_3 j_4|$$

$$\Rightarrow \left(\frac{1}{N}\right)^4 \frac{1}{d^2} \sum' \sum' \sum' \sum' \left(\omega^{j_1 k_1}\right)^* \left(\omega^{j_2 k_2}\right)^* \left(\omega^{j_3 k_3}\right)^* \left(\omega^{j_4 k_4}\right)^*$$

$$|k_1' k_2' k_3' k_4'\rangle\langle j_1' j_2' j_3' j_4'| \left(\sum_{P}' P^{\otimes 4}\right)$$

$$\sum' \sum' \sum' \sum' \omega^{j_1 k_1} \omega^{j_2 k_2} \omega^{j_3 k_3} \omega^{j_4 k_4} \, |k_2 k_1 k_3 k_4\rangle\langle j_1 j_2 j_3 j_4|$$

$$\omega^{jk}(N) = e^{2\pi i \frac{jk}{N}}$$

GIANLUCA:

$$M_{LIN}(U) = 1 - d\, \mathbb{E}_c \left[\mathrm{Tr}\left(Q\, U^{\otimes 4} C^{\otimes 4} \psi^{\otimes 4} C^{+\otimes} U^{+\otimes 4} \right) \right]$$

$$= 1 - d \Big(\alpha\, \mathrm{Tr}\left[Q\, U^{\otimes 4}\, Q \Pi_{sym} (U^+)^{\otimes 4} \right] \quad 5/6$$

$$- \beta\, \mathrm{Tr}\left[Q U^{\otimes 4} \Pi_{sym} (U^+)^{\otimes 4} \right] \Big)$$

SIMPLER ONE

$$\mathrm{Tr}\left(Q\, U^{\otimes 4}\, \Pi_{sym} (U^+)^{\otimes 4} \right)$$

$$\frac{1}{d^6} \mathrm{Tr}\Big(\sum_P \sum_{\vec{k}\vec{j}}' \sum_{\vec{k}'\vec{j}'}' w^{\vec{j}\vec{k}}\, w^{\vec{j}'\vec{k}'}$$

$$|k_1' k_2' k_3' k_4'\rangle\langle j_1' j_2' j_3' j_4'|\, P^{\otimes 4}\, |k_1 k_2 k_3 k_4\rangle$$

$$\langle j_1 j_2 j_3 j_4|\, \Pi_{sym} \Big)$$

FOCUSING ON

$$\Rightarrow \langle j_1' j_2' j_3' j_4'|\, P^{\otimes 4}\, |k_1 k_2 k_3 k_4\rangle$$

$$= \langle j_1'|P|k_1\rangle \otimes \langle j_2'|P|k_2\rangle \otimes$$

$$\langle j_3'|P|k_3\rangle \otimes \langle j_4'|P|k_4\rangle$$

USING BIT STRINGS WE CAN EXPRESS PAULIES AS (GIANLUCA REF.)

$$P = \sum_{\{x\},\{z\}} z^z x^x$$

$x, z \in \{0,1\}^{\otimes n}$ BIT STRINGS

$$\Downarrow$$

$$\sum_{i,j} \langle j_1' | z^{z_i} x^{x_{ij}} | k_1 \rangle$$

ACTING ON BRA — ACTING ON KET

$$= \sum_{i,j} (-1)^{j_1} \langle j_1' | x \oplus k_1 \rangle$$

ADDITION MOD 2 OPERATION

$$= (-1)^{j_1' \cdot z} \delta_{j_1, k_1 \oplus x}$$